

SECURE INFORMATION SYSTEM: A MODIFIED ALGORITHM FOR ADVANCED ENCRYPTION STANDARDS

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Abstract - In this research paper we examine the need for a solution to the currently existing concern areas within any organization, pertaining to: secure processing of information within the organization, efficient administration ,control , and enhanced work efficiency within an organization Enterprise employees are no longer just working individually, but in a distributed environment, demanding seamless access to the organization's crucial data resources from various departments requiring a network communication for data transfer, sharing, database access etc. demanding a secure, trustworthy and efficient networked environment from an organization's security point of view and also with the point of view with respect to the efficiency of an organization.

Keywords: Network security, Administration, SSL, Performance analysis, Secure data sharing, encryption/ decryption, RBAC, Authentication, AES.

I. INTRODUCTION

Security and network level authentication are one of the fundamental problems that most organizations are currently facing both dealing within the organization processes and intra-organization processes. Security is a primary reason for the deployment of authentication systems. Security is also vital to the preservation of privacy in that one must make use of security technology in order to protect privacy-related information. It is not simply the technical mechanisms of security that matter but also the processes and policies that govern the access to sensitive data. It is therefore essential to understand the security requirements and properties of both the authentication system itself and the resources it is protecting. As a lot of information assets are usually dependent and processed over a networked environment system (especially **enterprise information system**) dealing in critical processes like Data Transfer, Data Access, Client-Client interactions, and Information Sharing which raise the security concerns like data integrity, secured data

transfer medium, encrypted information, authentication etc[4-8]. Any compromise in the security of information may cause great losses to users and organizations.

The security goals of most enterprise-wide application systems include confidentiality, integrity, availability and accountability. To achieve these goals, enterprise-wide network security tools and a secure network communication system needs to be developed to meet security requirements of such organizations. Notion of user-centered design is being proposed, with particular emphasis on the authentication context, an integrated solution comprising of security and administration is discussed with remaining future extensions therefore essential to understand the security[4-8].

The idea of multi-level security has been used in many aspects such as for protecting relational database network objects, for managing the evolution of web applications etc[8]. We plan to develop a multi-level security version for Information Systems which allows secure processing of information within the organization, efficient administration, control, and enhanced work efficiency within an organization. The current direction for this design is to use and implement a five level security structure including RBAC to provide enterprise-wide network security tools and a secure communication network needed by most of the organizations. This information system is called Secure Information System. ESCORT is an Enterprise, policy-based security protocol for protecting relational database network objects [3]. ESCORT provides an end-to-end security architecture maintaining Confidentiality and integrity of database objects flowing over network links within the (EIS) Enterprise Information System. The second is the DOFS architecture [4-8], which prevents users, authorized or not, from having access to the physical bits of sensitive files, and thus provides a different, presumably stronger protection against information theft committed by corporate insiders than most existing enterprise. DOFS provides this protection

without affecting enterprise users' normal working practice and it's completely compatible with existing IT infrastructures. Thus we developed different type of securities that is required at organizational level. We can categories them into 3 different parts Browser Level, System Level and Database Level. At Browser Level we can use SSL security in order to prevent it from external threats. System Level security can be provided by using operating system security features comprising of access policy control and group policy methods that can be deployed manually. At database level we can provide security by integrating by integrating browser level security features and inbuilt features provided by the database software[7-8].

II. BACKGROUND

In any organization it is the management which is responsible for the reliable and secure operation of the information systems that supports the organization. As inter and intra-department networking between information systems become the rule as well as the daily operational environment of an organization, the scope of concern takes on new aspects and new technical details come into play. The paper discusses the newly designed architecture and security features to provide a novel solution to the discussed problems and its implementation success providing various enterprise-wide processes and secure administration in a networked environment. The focus is to enable management teams and technicians to enhance control and ensure secure management of their responsibility in respect of the reliable and secure operation of the information networks which support their organizations, their customers or the general public.

A. Existing Security Models

There has been a lot of effort been put in regarding the research work in terms of security and privacy of database information at the storage level. This research mainly focuses on encrypting the data stored in the database. Although this can protect the data from a hacker breaking into the database server or from the network or domain administrators, but it doesn't protect the privacy or integrity of data traveling over the network links between the application client and the relational database[8]. At the network level, SSL is the standard security protocol for providing confidentiality and integrity services for network data on the Internet. It provides point-to-point security by establishing a secure

channel on top of TCP, where it supports server authentication using certificates, confidentiality, and message integrity.

B. Need for Enterprise-wide Security Management for Administration

The technological advancement of the systems have posed many technical and economic challenges, especially when corporate downsizing practices must co-exist with the retention and integration of proven legacy IS systems. Enterprises have now discovered new opportunities by transferring their decentralized organizational structures to distributed information systems. Data which logically belonged together was distributed across several locations and managed by only one database software toolset but now heterogeneous distribution is today's center of interest[8]. Here we have locally independent subsystems communicating with each other in a structured way -- horizontally between Client/Server domains and vertically with centralized host systems. In parallel with this, technologically advanced industries have already positioned themselves to both the workplace transformation and the information society. Consequently, they now make portions of their applications sets accessible to very large user groups via open networks. This arises a critical need for Web based secure administration solution that not only provides network security, but also ensures availability of better administration tool set providing various needed features including information sharing, data transfer, inter-communication, problem solving etc.

III. ARCHITECTURE

The Secure Enterprise Administration software product operates on large systems under Windows operating system, and java runtime environment. The software operates on the client and server sides by seamlessly employing a Security Engine component to transparently control the database network data encryption/decryption operations. Software architecture that is especially useful for managing the evolution of web applications. Web server systems are often organized as a collection of tiers that divide responsibilities between specialized software components. In a typical case, a system is organized as an application server that is attached to the Internet and communicates with back-end components such as a database. When users require service, they authenticate to the application server, which determines their authorizations and uses these to modify the database according to the privileges of the principal that represents the user on the system.

Although the application server in such systems is acting on behalf of a single principal in this instant, it requires general access to the database because it may act on behalf of many different principals. This is worrisome in the face of threats like identity theft because an attacker who manages to compromise the application server may be able to execute a wide range of actions relative to the database, such as querying for he names and Social Security numbers of all principals in the system.

Architectural Pattern

The Model-View-Controller architectural pattern (MVC) divides an interactive application into three components. The model given in Fig. 1 contains the core functionality and data[7]. Views display information to the user. Controllers handle user input. Views and controllers together comprise the user interface. A change-propagation mechanism ensures consistency between the user interface and the model.

Successful use of the pattern isolates business logic from user interface considerations, resulting in an application where it is easier to modify either the visual appearance of the application or the underlying business rules without affecting the other.

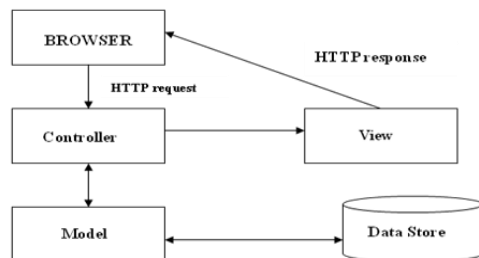


Fig. 1 MVC Pattern

IV. SECURITY IN LAYERS

Before getting into the specifics of IPsec, let's step back and consider what it means to provide security at the network layer. Consider first what it means to provide network layer secrecy. The network layer would provide secrecy if all data carried by all IP datagrams were encrypted. This means that whenever a host wants to send a datagram, it encrypts the data field of the datagram before shipping it out into the network[10].

A. Secure Socket Layer(SSL)

Secure sockets layer (SSL), originally developed by Netscape, is a protocol designed to provide data encryption and authentication between a Web client and a Web server. The protocol begins with a handshake phase that negotiates an encryption algorithm (e.g., DES or RSA) and keys, and authenticates the server to the client. Optionally, the client can also be authenticated to the server. Once the handshake is complete and the transmission of application data begins, and all data is encrypted using session keys negotiated during the handshake phase. SSL provides the following features:

SSL server authentication, allowing a user to confirm a server's identity.

SSL client authentication, allowing a server to confirm a user's identity. Analogous to server authentication

An encrypted SSL session, in which all information sent between browser and server is encrypted by sending software (browser or Web server) and decrypted by the receiving software (browser or Web server)..

B. Transport Layer Security

TLS (Transport Layer Security) is a protocol for establishing a secure connection between a client and a server[10]. TLS is capable of authenticating both the client and the server and creating a encrypted connection between the two

C. Network Layer Security

In Transport Mode, IPsec encrypts the payload of each IP packet, but not the packet headers. In Tunnel Mode, IPsec encrypts the payload and the headers of each IP packet. Many networks which are not able to support Tunnel Mode are able to successfully support Transport

V. MODIFIED AES ALGORITHM

Advanced Encryption Dtandards (AES) is used to apply a high level of security. In this section, we proposed own algorithm for security by modifying the standard functioning of AES as shown in Fig. 2. We have increased the security level but complexity will also increase asymptotically. In this algorithm, we consider 128 bit as input and 256 different keys of size 56 bit. Each level required one diffener key. At

the last iteration, we replaced L_{256} and LM_{256} to each other and RM_{256} and R_{256} to each other. Each L_j , R_j , LM_j and RM_j of 32 bit. It can be determined recursively as follows:

$$\begin{aligned} L_j &= LM_{j-1} \\ LM_j &= L_{j-1} \text{ XOR } K_{j-1} \\ RM_j &= R_j \\ R_j &= RM_{j-1} \text{ XOR } K_{j-1} \end{aligned}$$

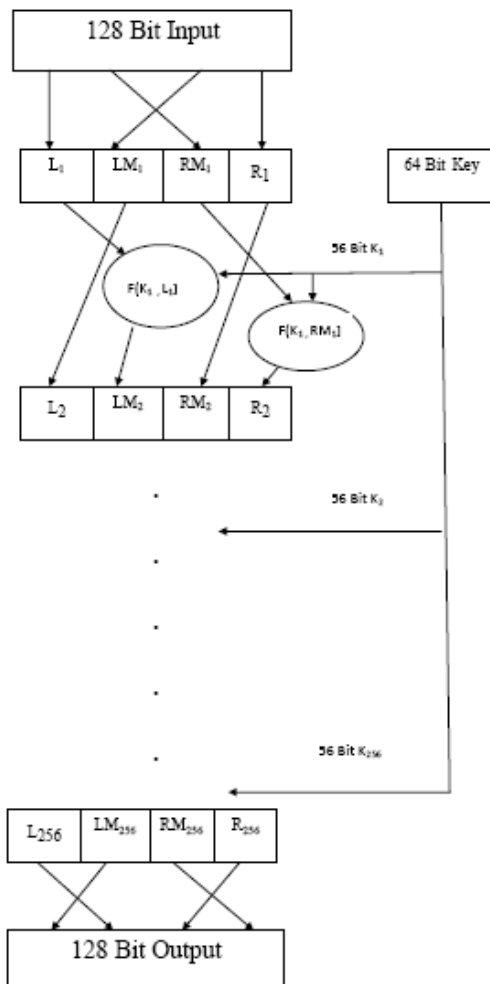


Fig. 2 128 bit Modified AES Algorithm

VI. Future Work and Conclusion

A better architecture for efficient and secure working, enterprise providing a secure desktop application providing basic features of intercommunication and information sharing along with employee administration was suggested. The suggested MVC pattern will improve the efficiency of the software and also make it very easy for the software to be updated. The design discussed can be implemented

for use in internal networks of various organizations and public domains, including the implementation of RBAC which will categorize the users on the basis of the role they are playing in the organization. The security needs of most enterprise-wide application systems: confidentiality, integrity, availability and accountability, etc are all catered to and successfully implemented and tested for practical use also providing enterprise-wide network security tools at different levels and a secure network communication system that are greatly needed to meet security requirements of most organizations. In this paper we proposed a new algorithm based on AES to increase the level of security.

Future scope of this project lies in the direction requiring work with collaboration, benchmark, and case-study partners to identify specific implementations of effective programs where security is being addressed from both enterprise and governance perspectives; capture key principles, policies, procedures, processes, practices, and measures; document these with attribution where appropriate, deriving and recommending strategies that will help organizations develop their own approaches for enterprise-security governance. Interoperable system compatibility, OS-independence, and further security functionality such as adding hardware security component can be seen as good extensions to this work. We can implement this modified AES algorithm to establish a high level of security for any data which are available on internet. It may be difficult to implement but if it has been implemented at once, it is sure that encrypted data will have high level of security.

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